

3 (a) (i) Any difference about Amplitude/Phase difference/Energy B1

Comparison between a stationary wave and a progressive wave

	Stationary	Progressive
Energy	No net transfer of energy from one point to another. Energy is confined within the wave and there is interchange of K.E. and P.E.	Energy is transferred in the direction of travel of the wave.
Phase	All particles between two adjacent nodes are <i>in phase</i> . Particles on opposite sides of a node will be in <i>anti-phase</i> .	All points within one wavelength vibrate with different phase.
Amplitude	Varies from zero at nodes to maximum at antinode.	Same for all particles in the wave.
Wavelength	2 x distance between adjacent nodes or antinodes	Distance between adjacent particles which are in phase.
Frequency	All particles vibrate in SHM with same frequency except at nodes	All points vibrate in SHM with same frequency.
Waveform	Does not advance	Advances in the direction of velocity of the wave

(b) (i) only transverse waves can be polarised B1

(i) the waves of equal f and amplitude in opposite direction interfere/superpose producing a stationary wave B1

stationary wave has nodes and antinodes B1

the resultant signal is zero at a node distance from max (antinode) B1

to zero (node) is $\lambda/4 = 0.75 \text{ cm}$ B1

(ii) emitted waves are polarised (in vertical plane) B1

when T_2 is rotated by 90° , the two waves at right angles superposed to produce a resultant constant amplitude (of $A\sqrt{2}$) B1

Since intensity is proportional to square of amplitude, signal intensity is halved) B1

Note: no stationary wave is produced since waves are at right angles to each other)

4 (a) n.d. across one lamp = 2.5 V

C2